

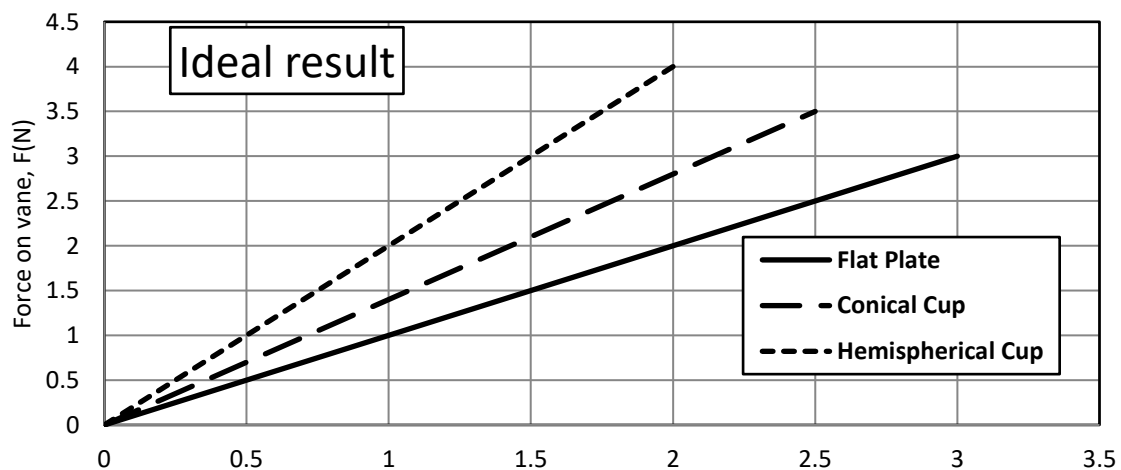
Exp.
05**Study of impact of Jet****Formulas and Constants:**

Density of water, ρ	=1000 kg/m ³
Diameter of the nozzle	=10mm
Cross sectional area of nozzle	=78.5mm ²
Mass of jockey weight	=600gm
Distance from center of vane to pivot of the lever	=0.15 m
Height of vane above tip of nozzle, s	= 35 mm
The jockey's weight weighs	$W=mg = (0.6 \times g) \text{ N.}$

- $F_{\text{exp}} = 4gy \text{ (N)}$
- $F_{\text{th}} = \dot{m}u_0(1 - \cos\beta)$
- $u = \frac{\dot{m}}{10^3 \times 78.5 \times 10^{-6}} = 12.75 \dot{m}$
- $u_0^2 = u^2 - 0.687$

Data Sheet

Deflector Shape:					Deflector Angle, β :				
No	Quantity (kg)	Time (s)	y (m)	$\dot{m}$ (kg/s)	u (m/s)	u_0 (m/s)	$\dot{m}u_0$ (N)	F_{exp} (N)	F_{th} (N)
01									
02									
03									
04									
05									
06									

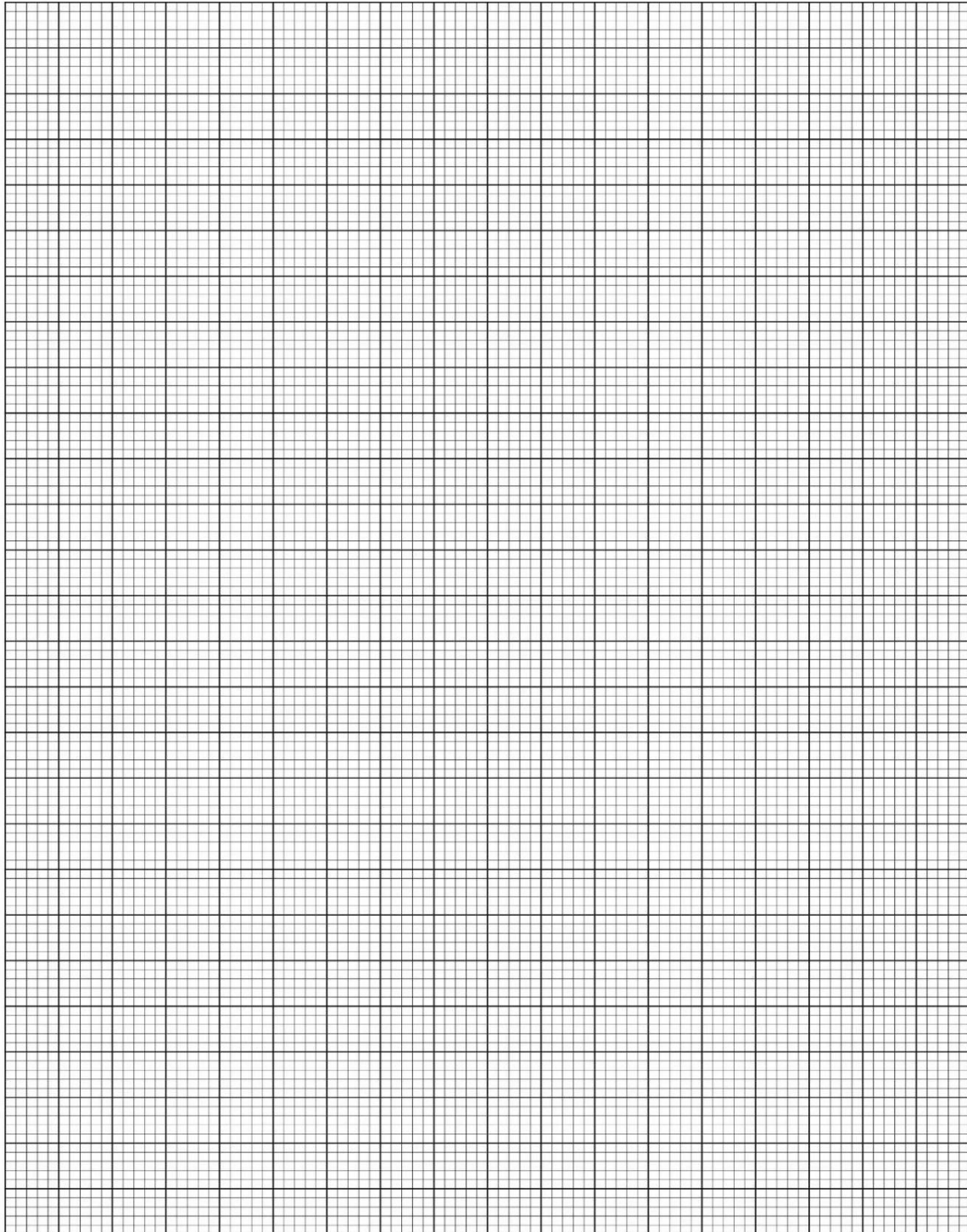


Plot:

- Plot F_{exp} vs μ_0 and F_{th} vs μ_0

Along Horizontal Axis:

Along Vertical Axis:



Sample Calculation:

- Show the sample calculation for just one of the observations.
- Find the slope of the F_{exp} vs μ_0 graph and compare the values for slope of curve with the ideal data. [**Ideal result for Flat plate: 1.0, Cone: 1.5, Hemisphere: 2.0**]

Discussion:

- Compare the theoretical and experimental forces obtained for the respected vane. Discuss the reasons for any differences.
- If the experiment were repeated using a different nozzle diameter, what changes would you expect in the results?
- Suggest at least three improvements to the experimental procedure that would reduce measurement errors and improve repeatability.